

AITIR Future Development Plan and Prong 3 Addendum

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Proposed Endeavor: AI-Assisted Adaptive Threat Intelligence and Response Framework for Public-Sector and Law-Enforcement Cybersecurity

Purpose of This Addendum

This addendum strengthens the third prong of Matter of Dhanasar by explaining how Mr. Hossain will continue advancing the AI-Assisted Adaptive Threat Intelligence and Response Framework, known as AITIR, beyond the needs of any single employer. The plan covers the next 12 to 24 months and identifies specific research, publication, framework-development, public dissemination, and professional-outreach activities.

The plan is designed to show that AITIR is not merely part of Mr. Hossain's current employment. It is a portable public-sector cybersecurity endeavor that can continue through research, publication, technical documentation, professional service, and outreach to public-sector cybersecurity practitioners.

Summary of the Future Plan

Over the next 12 to 24 months, Mr. Hossain will continue developing AITIR through five coordinated workstreams: technical framework refinement, peer-reviewed publication, public cybersecurity writing, framework/model release, and public-sector cybersecurity outreach.

- Technical framework refinement: further define AITIR inputs, risk indicators, identity-centered threat patterns, prioritization logic, and response workflow.
- Planned peer-reviewed publications: prepare additional manuscripts on AI-assisted identity threat detection, public-sector cyber resilience, and adaptive response models.
- Public cybersecurity articles: publish practitioner-facing articles explaining identity-centered cyber risk and AITIR concepts for a broader cybersecurity audience.
- Framework/model release: prepare a practical AITIR model document, architecture diagram, and implementation guide suitable for public-sector cybersecurity review.
- Outreach: engage cybersecurity professionals through professional associations, peer review, editorial service, and targeted communication with public-sector cybersecurity practitioners.

12 to 24 Month Development Roadmap

Phase I: Months 1-6 - Framework Refinement and Technical Documentation

Mr. Hossain will refine the core AITIR architecture by defining its operational components: identity and access activity, authentication behavior, privileged-access indicators, endpoint and remote-access context, threat intelligence inputs, risk scoring, and response-prioritization workflow. This phase will convert the AITIR concept into a clearer technical model that can be evaluated by researchers and practitioners.

- Deliverable 1: AITIR architecture and operational model document.

- Deliverable 2: AITIR threat-to-response workflow showing how identity events become risk-prioritized response actions.
- Deliverable 3: short technical brief comparing static access-control review with adaptive identity-threat analysis.

This phase supports Prong 3 because it demonstrates that the endeavor can proceed through independent framework development and technical documentation, not only through a single employer-sponsored role.

Phase II: Months 4-12 - Peer-Reviewed Research and Scholarly Dissemination

Mr. Hossain will prepare and submit additional scholarly work focused on public-sector cybersecurity resilience and AI-assisted identity threat detection. The publications will continue the research trajectory already started through his AITIR-related publication and related work in cybersecurity, AI-assisted defense, post-quantum cryptography, and secure systems.

- Planned manuscript topic 1: adaptive identity-threat detection for public-sector cybersecurity environments.
- Planned manuscript topic 2: AITIR-based risk prioritization and response support for law-enforcement and government systems.
- Planned manuscript topic 3: integration of AITIR concepts with NIST SP 800-53, NIST Cybersecurity Framework 2.0, zero trust, and incident-response practices.

This phase supports Prong 3 because peer-reviewed dissemination is not employer-specific. It allows other cybersecurity researchers and public-sector practitioners to evaluate, critique, and build on the framework.

Phase III: Months 6-18 - Public Cybersecurity Articles and Practitioner Materials

Mr. Hossain will also publish practitioner-facing cybersecurity articles explaining AITIR concepts in plain language for operational audiences. These articles will focus on identity-centered cyber risk, access misuse, privileged-account monitoring, adaptive threat intelligence, and public-sector cyber resilience.

- Planned article topic 1: why identity and access activity should be treated as threat intelligence in public-sector systems.
- Planned article topic 2: practical challenges in protecting law-enforcement and government systems from credential misuse and access-related threats.
- Planned article topic 3: how adaptive threat intelligence can support incident response and operational continuity.

This phase supports Prong 3 because public writing broadens the reach of the endeavor beyond academic journals and beyond Mr. Hossain's current employer. It creates a public dissemination path for cybersecurity professionals who may not read scholarly publications.

Phase IV: Months 12-24 - AITIR Framework/Model Release

Mr. Hossain will prepare a consolidated AITIR framework/model release. This release will not disclose sensitive employer systems or confidential operational details. It will provide a generalized model that public-sector cybersecurity professionals can review and adapt to their own environments.

- Release component 1: AITIR framework overview and use-case narrative.
- Release component 2: identity threat indicator categories and prioritization logic.
- Release component 3: response workflow mapping identity alerts to triage, escalation, containment, audit, and continuity considerations.
- Release component 4: alignment notes connecting AITIR concepts to NIST SP 800-53 control families, NIST Cybersecurity Framework 2.0 functions, zero trust concepts, and incident-response practices.

This phase directly supports the waiver request. A public framework/model release is field infrastructure. It is broader than a job assignment and is not captured by a PERM labor-market test for one employer.

Phase V: Months 12-24 - Outreach to Public-Sector Cybersecurity Professionals

Mr. Hossain will use professional channels to share AITIR concepts with cybersecurity professionals who work in or support public-sector environments. The intended audience includes law-enforcement technology professionals, state and local government cybersecurity personnel, identity/access management practitioners, GRC/RMF professionals, incident-response professionals, and researchers studying AI-assisted cybersecurity.

- Professional outreach through IEEE, IEEE Computer Society, ISSA, IAENG, peer-review activities, editorial engagement, and cybersecurity professional networks.
- Targeted sharing of non-confidential AITIR materials with public-sector cybersecurity professionals and independent experts willing to review or critique the framework.
- Continued peer-review and editorial service to stay connected with emerging research in cybersecurity, AI-assisted defense, identity security, and public-sector cyber resilience.

This phase supports Prong 3 because it shows a credible pathway for public benefit beyond a single employer. The outreach plan is designed to help AITIR become a transferable cybersecurity framework rather than an internal workplace project.

Why the Endeavor Is Independent from One Employer

Mr. Hossain's current employment at the Kansas Bureau of Investigation is strong evidence that he is well positioned to understand public-sector cybersecurity problems. However, the proposed endeavor is not limited to KBI. AITIR can be advanced through activities that do not depend on any one employer: technical documentation, scholarly publications, public cybersecurity articles, framework/model release, peer review, editorial service, professional outreach, and continued independent research.

The core value of AITIR is the development and dissemination of a cybersecurity framework for identity-centered threat detection and adaptive response in public-sector environments. A framework can be studied, refined, published, reviewed, and adapted across agencies and institutions. That structure makes the endeavor portable and broader than one job description.

For this reason, the labor certification process is not well suited to evaluate the full national value of the endeavor. PERM would test whether a single employer has a specific job available and whether qualified U.S. workers are available for that job. It would not evaluate Mr. Hossain's broader

framework-development, publication, dissemination, and public-sector cybersecurity outreach activities.

Prong 3 Argument Supported by This Plan

This future plan strengthens the third Dhanasar prong for three reasons.

- First, it shows that the endeavor is active and concrete. Mr. Hossain has identified specific deliverables over the next 12 to 24 months, including framework documentation, planned publications, public cybersecurity articles, and an AITIR model release.
- Second, it shows that the endeavor extends beyond any single employer. The planned work can proceed through research, publication, professional service, and public dissemination regardless of one employer's internal priorities.
- Third, it shows that the United States benefits from waiver even assuming other qualified cybersecurity workers exist. The benefit is not based on generalized occupational demand. It is based on continuation of Mr. Hossain's specific AITIR framework and his combination of public-sector technology experience, law-enforcement technology exposure, cybersecurity research, and professional engagement.

Accordingly, the national interest served by Mr. Hossain's continued advancement of AITIR outweighs the national interest inherent in requiring a job offer and labor certification. The waiver would allow him to continue developing and disseminating a public-sector cybersecurity framework addressing identity-centered threats, adaptive threat intelligence, risk prioritization, and resilient response for government and law-enforcement environments.

Suggested Integration into the Petition Letter

This addendum can be cited in the petition letter under Dhanasar Prong 3 and in the exhibit index as part of Exhibit 11, AITIR Framework Architecture and Operational Model. It should be cross-referenced in the sections discussing future plan, portability beyond KBI, and why labor certification is structurally inadequate for this endeavor.

Suggested petition sentence: "Mr. Hossain has prepared a 12 to 24 month AITIR Future Development Plan identifying specific framework, publication, public-writing, model-release, and outreach deliverables, demonstrating that the endeavor will continue through portable research and professional dissemination beyond any single employer."